

Structured Subsystem-Aware Feature Representations for Predictive Maintenance in Metro Systems

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- Metro systems generate large-scale multivariate sensor data
- Service-affecting failures (SFAs) reduce availability and reliability
- Predictive maintenance aims to detect degradation early

Challenges in Real-World Deployment

- Rare and imbalanced failure events
- High false-alarm rates
- Limited generalization on unseen operational conditions
- Lack of subsystem-aware feature organization

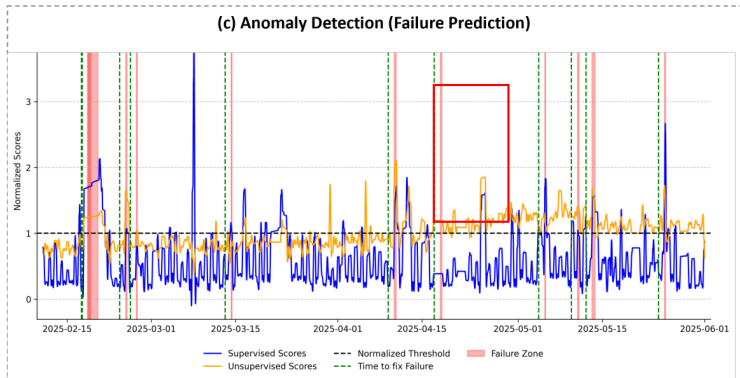
- Existing approaches process sensor data as one high-dimensional feature space
- Functional subsystem relationships are often ignored
- This may introduce:
 - Noise amplification
 - Reduced interpretability
 - High false-alarm rates
 - Poor generalization

Research Gap

The influence of subsystem-based structured representations on predictive maintenance performance remains largely unexplored.

[1] Steiner et al., IEEE RAMS, 2026.

Real-World Prediction Challenges



Supervised and unsupervised anomaly scores on the MetroAT dataset. The highlighted regions illustrate false alarms occurring outside actual failure intervals.

Adapted from Steiner et al., IEEE RAMS 2026.

Related Work and Research Gap

- **Time-Series Learning**

- LSTM, GRU, and TCN are widely used for multivariate time-series modeling [1,2]
- Machine learning and deep learning approaches have demonstrated strong performance for multivariate time-series modeling [3]

- **Findings from MetroAT**

- Limited generalization on unseen operational data [4]
- High false-alarm rates in real-world deployment settings [4]

- **Feature Representation Challenges**

- Sensor variables commonly treated as a flat high-dimensional feature space [5]
- Limited investigation of subsystem-aware feature organization [5]

- **Research Gap**

- The influence of subsystem-aware feature representations on predictive maintenance performance, robustness, and generalization remains largely unexplored [4,5]

Proposed Approach

Subsystem-aware feature representations provide more coherent information than a single high-dimensional feature space, leading to improved predictive maintenance performance.

Approach

- Analyze anomaly-centered sensor windows
- Discover recurring anomaly patterns using clustering and temporal pattern analysis, and verify the identified anomaly groups with domain experts
- Transform the verified anomaly groups into subsystem-aware feature representations and construct subsystem-specific datasets
- Compare against conventional full-feature representations

Expected Outcome

- Reduced false alarms, Improved robustness
- Better generalization performance

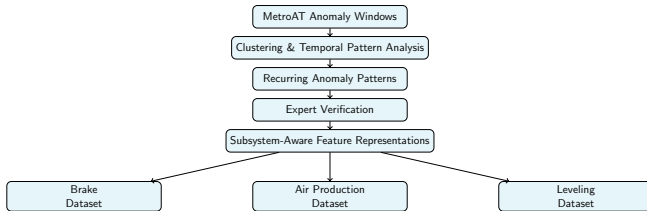
RQ1

To what extent can subsystem-aware feature representations constructed from recurring anomaly patterns extracted from MetroAT anomaly windows produce subsystem structures that are verified by domain experts?

RQ2

To what extent do subsystem-aware feature representations improve predictive maintenance performance compared with conventional full-feature representations under identical experimental conditions?

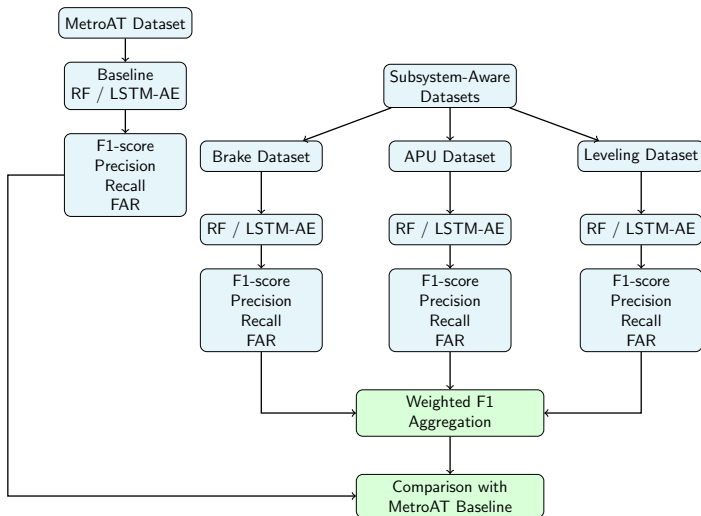
Proposed Research Workflow



Output:

Construction of subsystem-aware feature representations by discovering recurring anomaly patterns through clustering and temporal pattern analysis, followed by expert verification.

Evaluation Framework



Baseline: MetroAT evaluated using RF and LSTM-AE.

Proposed: Brake, APU, and Leveling datasets evaluated under identical settings.

Evaluation: Subsystem F1-scores are aggregated and compared with the MetroAT baseline; Precision, Recall, and FAR are reported separately.

Example Comparison Table

Representation	F1-score	Precision	Recall	FAR
MetroAT Baseline	–	–	–	–
Brake Dataset	–	–	–	–
Air Production Dataset	–	–	–	–
Leveling Dataset	–	–	–	–
Subsystem-Aware (Weighted F1)	–	–	–	–

MetroAT serves as the baseline representation.

Brake, Air Production, and Leveling datasets are evaluated independently using identical models and experimental settings.

The subsystem F1-scores are aggregated into a weighted overall F1-score and compared against the MetroAT baseline.

Precision, Recall, and FAR are retained at subsystem level to provide diagnostic insights into subsystem-specific performance.

Framework: CRISP-DM

- Business and data understanding
- Unsupervised pattern discovery and subsystem-aware representation construction
- Temporal modeling approaches
- Robustness and performance evaluation

Experimental Design

- Baseline: Unstructured representation
- Comparison: Structured vs. unstructured representations
- Controlled setup:
 - Same *temporal splits, hyperparameters, evaluation pipeline*

Expert Verification

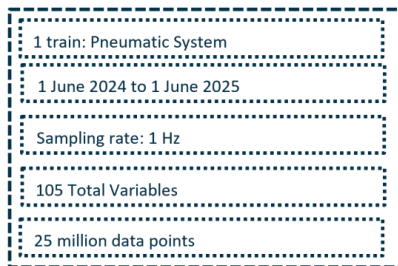
- Expert verification of anomaly groups
- Verification of subsystem assignments

Structured vs. Unstructured Representation

Unstructured Representation	Subsystem-Aware Representation
All sensor variables used jointly	Sensor variables grouped using clustering and temporal pattern analysis
Flat high-dimensional feature space	Subsystem-aware feature representations
Single representation	Multiple verified subsystem-specific datasets
Potential noise amplification	Reduced subsystem-level noise influence
Limited subsystem interpretability	Subsystem-specific diagnostic insights

Goal: Evaluate whether subsystem-aware feature representations improve predictive performance, robustness, and generalization compared to full-feature representations.

Experimental Setup: MetroAT Dataset



MetroAT dataset adapted from Steiner et al., CIRP CMS 2025.

Subsystem Coverage

- Air Production Unit (APU)
- Pneumatic braking subsystem
- Leveling subsystem

Dataset Characteristics

- Event-aware operational dataset
- Real-world metro operations
- High-dimensional multivariate time-series
- Rare failure events

Evaluation Metrics

- F1-score (primary comparison metric)
- Precision, Recall, and False Alarm Rate (FAR)
- Weighted aggregation of subsystem F1-scores

Validation Strategy

- Temporal train-test split
- Identical models and hyperparameters
- Controlled comparison of structured and unstructured representations

Evaluation Goal

- Compare subsystem-aware feature representations with the MetroAT baseline representation
- Assess subsystem-specific predictive contribution
- Evaluate robustness and generalization performance

Thank You!

Questions & Discussion